

Aram Ebtekar

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Employment History

- 2026  **AIXI Labs, Founding Research Scientist.** Leading the research agenda of a >\$3M non-profit on learning theory and agency, building on Marcus Hutter's UAI/AIXI framework; mentoring and co-supervising research fellows; forming external collaborations.
- 2025 – 2026  **MATS + UC Berkeley, Research Fellow.** Advanced the mathematics of safe reinforcement learning agents with Michael Cohen; reviewed papers for AGI-25, NeurIPS and Nature; co-organized the international reading group on Universal Algorithmic Intelligence.
- 2025  **Anthropic, AI Safety Research Fellow.** Co-developed the inoculation prompting technique to mitigate reward hacking in Large Language Models.
- 2021 – 2024  **Independent, Researcher.** Published very cool papers on foundations of physics and causality; took contracts deploying my Elo-MMR algorithm to international competitions, and implementing planning algorithms for autonomous Caterpillar vehicles.
- 2018 – 2019  **Mythic, Senior AI Research Scientist.** Led two exploratory efforts: video super-resolution, covering the whole pipeline from dataset collection to neural architecture design; and hardware co-design investigations on how efficiently convolutional layers map to analog hardware, and how long the resulting models retain their accuracy.
- 2016 – 2018  **Waymo, Behavior Prediction Research Engineer.** Framed driver behavior as mathematical optimization problems with efficient algorithmic solutions, enabling autonomous cars to predict surrounding drivers' movements in real-time; and led a study group to develop expertise in deep reinforcement learning techniques.

Education

- 2019 – 2022  **Self-directed studies.** An eclectic mix of graduate-level math, physics, economics, statistics, information theory, and undergrad-level humanities, gathered from a combination of UBC courses, MIT OpenCourseWare, textbooks, papers, and more, as preparation for deeply technical interdisciplinary research.
- 2012 – 2015  **Carnegie Mellon University, M.S. (Ph.D. candidate) in Computer Science.**
 - NSERC Canada Graduate Scholarship M
 - Research projects in hybrid systems verification and search-based planning
 - Completed the 2012 Summer School in Algorithmic Economics
 - Memberships: Graduate Student Assembly departmental representatives, Ballroom Dance Club, School of Computer Science musical performances
- 2008 – 2012  **University of British Columbia, B.Sc. Honours in Computer Science & Mathematics.**
 - Research projects in evolutionary game theory and computational geometry
 - GPA: 92% (A+)

Research Publications

- 1 A. Ebtekar and M. K. Cohen, "Golden handcuffs make safer AI agents," *in preparation for ALT 2027*, arXiv: 2604.13609 [cs.LG].
- 2 A. Ebtekar, M. Hutter, and D. J. Sutherland, "Understanding generalization requires universal induction," *submitted to NeurIPS position track*, 2026.  URL: <https://djsutherland.ml/papers/nfl-ait.pdf>.
- 3 N. Wichers, A. Ebtekar, A. Azarbal, *et al.*, "Inoculation prompting: Instructing LLMs to misbehave at train-time improves test-time alignment," 2026. arXiv: 2510.05024 [cs.LG].

- 4 A. Ebtekar and M. Hutter, “Foundations of algorithmic thermodynamics,” *Physical Review E*, 2025.  DOI: 10.1103/PhysRevE.111.014118.
- 5 A. Ebtekar, Y. Wang, and D. Janzing, “Toward universal laws of outlier propagation,” *Conference on Uncertainty In Artificial Intelligence*, 2025.  URL: <https://dl.acm.org/doi/10.5555/3762387.3762440>.
- 6 A. Ebtekar and M. Hutter, “Modeling the arrows of time with causal multibaker maps,” *Entropy cover article (top 1%), Editor’s Choice*, vol. 26, no. 9, p. 776, 2024.  DOI: 10.3390/e26090776.
- 7 A. Ebtekar and P. Liu, “Elo-MMR: A rating system for massive multiplayer competitions,” in *Proceedings of The Web Conference*, 2021, pp. 1772–1784.  DOI: 10.1145/3442381.3450091.

Skills

Nat. Languages	📖	Fluent in English, French, Persian. Beginner in Spanish, Mandarin, Japanese.
Prog. Languages	📖	Rust, C++17, $\text{\LaTeX}$, Python, PyTorch, Keras, Java, C.
Academic	📖	General cs/math/stats/physics/econ, algorithmic information theory, statistical learning theory, agent foundations, robotics, reinforcement learning, software design, experiment design, technical writing, teaching, interdisciplinary research.

Miscellaneous Experience

Contest Achievements

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| 2015 | 📖 | 61st place globally among over 50,000 registrants in the Google Code Jam. |
| | 📖 | 57th place globally in the Topcoder Open Algorithm Competition. |
| | 📖 | 6th place in the North American Invitational Programming Contest’s Open Division, despite competing solo against teams of up to three. |
| | 📖 | Codeforces Grandmaster, former 1st in Canada, peak rating 2400+ on Codeforces and Topcoder. |
| | 📖 | ACM ICPC Pacific Northwest regional contest problem setter, author of problems J,L,N. |
| 2012 | 📖 | 18th place in the ACM ICPC World Finals in Warsaw, Poland. |
| 2011 | 📖 | Top 250, Team Honorable Mention in the William Lowell Putnam Mathematical Competition. |
| | 📖 | UBC Thunderbots team, 9th place in the RoboCup SSL international robot soccer competition. |

Selected Projects

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| 2020 | 📖 | Technical Blogging. My most popular article on foundations of probability made the front page of Hacker News and received over 200 comments. |
| 2017 | 📖 | Rust Algorithms Cookbook. A collection of classic algorithms and data structures elegantly crafted in Rust, serving as a proof of concept of the language’s compile-time safety discipline in contest programming. On 20/06/2017, it was the #1 trending GitHub repository globally. |
| 2012 | 📖 | U! Robot! Lead engineer in a team of 8 developers at a 48-hour Global Game Jam, completing a platforming game that was showcased in the top 3. |